

KOLAPALLI ADITYA SAI

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PROFILE SUMMARY

AI/ML Engineer with experience building production-oriented Generative AI, agentic AI, backend, automation, and computer vision systems. Skilled in Python, LLM orchestration, LangChain, LangGraph, FastAPI, Redis, PostgreSQL, and asynchronous distributed processing. Experienced in designing multi-tenant SaaS platforms, AI-powered analytics pipelines, autonomous research workflows, and real-time computer vision systems integrating LLM-based reasoning.

TECHNICAL SKILLS

AI/ML & GenAI	Python, LLMs, OpenAI API, LangChain, LangGraph, Agentic AI, RAG, Qdrant, Tool Calling, MCP, Prompt Engineering, Scikit-learn, PyTorch, Pandas, NumPy, YOLOv8, DeepSORT
LLM & Agent Infrastructure	Ollama, llama.cpp, Claude Code, Codex, OpenClaw, NemoClaw, Hermes Agent
Backend & Distributed Systems	FastAPI, REST APIs, Asynchronous Processing, Redis, Redis Queues, PostgreSQL, MinIO, Supabase, SQL, Neon DB, Firebase, Multi-Tenant Architecture, Rate Limiting, Retry Mechanisms, Idempotency
Programming	Python, Java, C, JavaScript, OOP, Data Structures & Algorithms, Linux
Automation & Developer Tools	n8n, Docker, Git, Playwright, Streamlit, JIRA, SMTP, API Integration
AI/Cloud Platforms	OpenAI, Anthropic, Google Cloud Vertex AI, Stability AI, Tavily

EXPERIENCE

AI Engineer <i>Kodryx AI</i>	Apr 2026 – Present Hyderabad
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- **Kodryx Social – Enterprise Messaging SaaS:** Engineered a high-concurrency, multi-tenant messaging platform using **FastAPI** for automated WhatsApp marketing campaigns. Designed distributed background processing with **Redis** queues and implemented rate limiting, idempotent operations, retry handling, and failure recovery for reliable bulk message delivery.
- Designed asynchronous campaign-processing workflows to decouple API requests from high-volume messaging operations, improving system resilience and preventing long-running campaign workloads from blocking application requests.
- **AI-Assisted Production Operations:** Deployed and securely configured **Hermes Agent** and **OpenClaw** on local servers used for application deployments, establishing controlled remote access for agent-assisted debugging, troubleshooting, and resolution of production issues. Enabled autonomous agents to analyze code, identify bugs, implement fixes, and support incident resolution with minimal manual intervention, resulting in a **30% improvement in development and operational efficiency**.
- **MEEAAROGYAM – AP Government RTGS Hackathon:** Contributed to the architecture and development of **MEEAAROGYAM**, an AI-driven healthcare platform featuring a **conversational voice agent** that enables users to book doctor appointments and schedule **ASHA worker visits entirely through phone calls**. Worked on the AI and backend integration powering voice-based interactions and automated appointment workflows, improving accessibility to healthcare services through conversational AI and contributing to the team's **1st Prize** win at the Andhra Pradesh Government RTGS Hackathon.

AI Engineer Intern <i>Kodryx AI</i>	Oct 2025 – Mar 2026 Hyderabad
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- **CultureSense – AI-Driven Employee Intelligence Platform:** Designed an end-to-end NLP and LLM pipeline using **LangChain** and **OpenAI** to analyze communication data from Teams, Slack, and Outlook. Built modular prompt workflows to transform unstructured conversations into structured sentiment, behavioral, engagement, and organizational-risk signals.
- Engineered AI-driven analytics workflows that aggregate communication signals into actionable organizational insights, enabling leadership teams to identify emerging trends in employee engagement and organizational health.
- **TANITA – Resilient Workflow Automation:** Developed a browser automation and report-distribution engine using **Playwright**. Implemented metadata-based matching to map dynamically generated PDF reports to user profiles and built state-persistent, resumable workflows capable of recovering from interrupted processing.

Content Moderator cum Data Research Analyst

Nov 2024 – Dec 2024

Concentrix – Avalon Project via Ixora Global Services

Hyderabad

- Processed and classified high-volume multimedia datasets for large-scale machine learning workflows, maintaining **99% data integrity** across annotation and validation tasks.

Artificial Intelligence Job Simulation Intern

Jun 2024

Cognizant via Forage

Remote

- Performed exploratory data analysis and developed machine learning models using Python to derive business-oriented insights from structured datasets.

PROJECTS

- **Autonomous LinkedIn Content Engine – n8n:** Designed an automated multi-node content generation and publishing workflow using **n8n**. Integrated **Tavily** for real-time research and **Stability AI** for visual asset generation. Developed custom JavaScript nodes for JSON sanitization, content transformation, and regex-based headline normalization.
- **CCTV AI Guard – Autonomous Computer Vision Surveillance:** Developed an end-to-end real-time vision pipeline using **YOLOv8** for human detection and **DeepSORT** for persistent multi-object tracking. Integrated **GPT-4o-mini** for LLM-based behavioral reasoning over tracking telemetry and implemented automated SMTP alerts with event snapshots for detected risks.
- **Phishing URL Detection System:** Developed a machine learning-based cybersecurity system achieving **96.9% accuracy** through URL feature engineering and ensemble classification techniques.

EDUCATION

B.Tech in Computer Science and Engineering – Artificial Intelligence & Machine Learning Jul 2025

Joginpally B R Engineering College, Hyderabad

CGPA: 7.89

PUBLICATIONS & CERTIFICATIONS

- **Research Publication:** “Fake Review Detection using Hybrid Deep Learning” – CNN + LSTM + Fuzzy Logic, published in the *European Advanced Journal*.
- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** – Oracle University
- **Claude Code in Action** – Anthropic
- **Introduction to Agent Skills** – Anthropic
- **Explore Generative AI with the Vertex AI Gemini API** – Google Cloud
- **Build Real World AI Applications with Gemini and Imagen** – Google Cloud
- **Prompt Design in Vertex AI** – Google Cloud
- **Computer Vision with Azure Cognitive Services** – Microsoft